

ASSIGNMENT - 03

1] Differentiate between single-layer and multi-layer neural network.

single layer neural network	Multi-layer neural network
1] A single-layer neural network consist of only one layer of weights connecting the input layer directly to the output layer	A multi-layer neural network consist of multiple layers, including one or more hidden layers between the input and output layer
2] In a single-layer neural network, there are no hidden layers present	In a multi-layer neural network one or more hidden layers are present to process information
3] A single-layer neural network can solve only linearly separable problem	A multi-layer can solve both linear and non-linear problem
4] This is simple in design and easy to implement	This is more complex due to multiple layers & connections
5] A single-layer has limited learning capacity	A multi-layer has high learning capability & can learn complex patterns

6] A single-layer neural network typically uses simple learning algorithms like back-rules such as the perceptron learning rule

7] A single-layer neural network requires less computation time

8] An example of perceptron is includes deep neural networks and feedforward neural network

The multi-layer neural network typically uses advanced algorithms like back-propagation for training

This requires more computation time due to multiple layers.

2] What is the role of an activation function in neural network? Explain with an example.

→ An activation function determines whether a neuron should be activated or not by transforming the input signal into an output signal. It introduces non-linearity into the neural network, which allows the model to learn and represent complex patterns in data. Without an activation function, a neural network would behave like a simple linear model, no matter how many layers it has.

Activation function also helps in controlling the output range, enabling better learning and stability during training. They play a crucial role in deciding how much information from a neuron is passed to the next layer.

Example: Consider a neuron that receives inputs and computes a weighted sum:

$$z = w_1x_1 + w_2x_2 + b$$

If we apply an activation function like the sigmoid function, the output becomes:

$$f(z) = \frac{1}{1 + e^{-z}}$$

The function converts any input value into a range between 0 and 1.

For eg:
If $z = 5$, output ≈ 0.99 (neuron strongly activated)

If $z = -5$, output ≈ 0.01 (neuron barely activated)

The activation function helps the neural network decide the importance of signals & enables it to learn complex relationships such as image recognition, speech processing, & classification tasks.

3] A perceptron has inputs $x = [2, -1, 1]$, weights $w = [0.5, 0.5, -0.5]$, and bias $b = 1$. Calculate &

(a) net input
(b) output using step activation function

Given &
Inputs: $x = [2, -1, 1]$
Weights: $w = [0.5, 0.5, -0.5]$
Bias: $b = 1$

(a) Net input we use the formula:

$$z = \sum_{i=1}^n w_i x_i + b$$

Now calculate &

$$z = (0.5 \times 2) + (0.5 \times -1) + (-0.5 \times 1) + 1$$

$$z = 0.5 - 0.5 - 0.5 + 1 = 0.5$$

net input = 0.5

(b) output using step Activation function

step function rule's

if $z \geq 0$, output = 1
if $z < 0$, output = 0

since $z = 0.5 > 0$
output = 1

4] consider a simple neural network with 2 input neurons, a hidden neuron and 1 output neuron

weights &
 Input $\rightarrow$ Hidden & $w_1 = 0.5$ $w_2 = 1$
 Hidden $\rightarrow$ output & $w_3 = 1$
 Bias = 0
 Input & $x = [1, 1]$

Tasks &
 a] compute the output of the hidden neuron using sigmoid activation given &

- Inputs & $x = [1, 1]$
- weights (Input $\rightarrow$ Hidden) & $w_1 = 0.5$
 $w_2 = 1$
- weight (Hidden $\rightarrow$ output) & $w_3 = 1$
- Bias = 0

a] first compute net input to hidden neuron &
 $z_h = (0.5 \times 1) + (1 \times 1) + 0 = 0.5$

Now apply sigmoid function

$$f(z) = \frac{1}{1 + e^{-z}}$$

$$\text{Hidden output} = \frac{1}{1 + e^{-0.5}} \approx 0.62$$

b] compute the final output of the network.

$\rightarrow$ Now input and output neuron &
 $z_o = (0.62 \times 1) + 0 = 0.62$

(Assuming sigmoid activation again)

$$\text{Final output} = \frac{1}{1 + e^{-0.62}} \approx 0.65$$

Hidden neuron output ≈ 0.62
 Final output ≈ 0.65

5] Train a perceptron to learn the XOR logic function. Use a two input perceptron model with bias and a step activation function. Show all steps including input-output mapping, initial weights, weight updates and final weights.

→ Step 1: XOR Truth table

x_1	x_2	Target (t)
0	0	0
0	1	1
1	0	1
1	1	0

Step 2: Initialize Parameters

- Initial weights: $w_1 = 0, w_2 = 0$
- Bias $b = 0$
- Learning rate: $\eta = 1$
- Activation: Step function

Step 3: Training Rule

- Net input: $z = w_1 x_1 + w_2 x_2 + b$
- $y = 1$ if $z > 0$, else 0

• weight update:

$$w_i = w_i + \eta(t - y)x_i$$

$$b = b + \eta(t - y)$$

Step 4: Training (3 Epoch Example)

x_1	x_2	t	z	y	Error (w_1, w_2)	b	
0	0	0	0	1	-1	(0,0)	-1
0	1	1	-1	0	+1	(0,1)	0
1	0	1	0	1	0	(0,1)	0
1	1	0	1	1	-1	(-1,0)	-1

Step 5: observation

After updating weights, if we continue training, the weights keep changing and no fixed solution is reached.

A two input single perceptron cannot learn the XOR function because XOR is non-linear problem. To solve XOR, we need a multi layer neural network (with hidden layer).